# When Euler Is Second-Order Accurate in Distribution: Global Criteria and Nonlinear Obstructions

Dev D. Goyal

***Abstract*—Explicit Euler can have second-order distribution accuracy while its particle trajectories remain first order. We characterize exactly when the maximum distribution error obeys a second-order bound for a fixed smooth field under global Lipschitz and second-moment assumptions. A transported-acceleration formula gives the sharp first-order Wasserstein coefficient of the recursively computed laws. Acceleration preservation is equivalent to superlinear maximum-node convergence, and also to superlinear terminal convergence for every smooth positive timestep density; otherwise the maximum-node error is first order. No prescribed finite collection of terminal schedule tests can certify this property. We then prove an obstruction applying to all globally Lipschitz nonlinear fields: in odd dimension, a scalar dilation of any finite nondegenerate Gaussian mixture admits acceleration preservation only for an affine scale, when the unique qualifying generator has Euler-exact trajectories. The proof combines a weighted stress identity with topology of remote density level sets. Gaussian specializations give explicit residual floors and necessary derivative growth. Convex Gaussian dilations pass every convex-observable energy test despite this obstruction. Nonradial positive constructions, residual certificates, and reproducible experiments complete the analysis. The results distinguish whole-path accuracy from terminal cancellation and linear-family failure from genuine nonlinear infeasibility.**

***Index Terms*—Flow matching, numerical integration, Wasserstein distance, probability transport, topological obstruction.**

**Impact statement.** Sampling systems are judged by output distributions, which can have different errors from individual trajectories. This work identifies when changing the velocity can improve Euler's distribution order and when every globally regular correction must fail. It also shows how endpoint-only convergence tests can miss first-order errors within the path. The experiments validate mathematical mechanisms; they establish no image-generation or deployment speedup.

## I. Introduction

Flow matching and probability-flow models generate samples by integrating a velocity field [1–3]. The numerical objective is often a distribution, even though a solver moves individual particles. Changing how particles are paired across time can therefore change numerical distribution error while preserving the exact marginal at every time. The relevant freedom is the established gauge condition

$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \qquad u_t = v_t + r_t, \qquad \nabla \cdot (p_t r_t) = 0.$$

It preserves the entire probability path, not just the endpoints [4–6].

For explicit Euler, the leading omitted trajectory displacement is one half the squared step size times the material acceleration $\mathbf{a}_t = \partial_t u_t + (D_x u_t) u_t$. If this acceleration also preserves the instantaneous distribution, its leading displacement can be removed by changing the coupling between numerical and exact outputs. Under global regularity and moment conditions this gives a finite-time $O(H^2)$ bound in full $W_2$, where $H$ is the largest step. Particle error can remain $\Theta(H)$.

Recent density-level backward-error analyses already identify the projected local defect and obtain sufficient order-elevation conditions [7, 8]. Two questions remain distinct from these local results. Is acceleration preservation also necessary for the actual recursively computed distribution path to converge superlinearly? And can a failure within linear or shellwise correction families be overcome by an arbitrary nonlinear field?

We answer the first question with an exact global Wasserstein expansion. For one fixed field, acceleration preservation is equivalent to maximum-node error $o(h)$ on uniform grids; it then gives $O(H^2)$ on every grid. Every other field has maximum-node error $\Theta(h)$. At a single terminal time, cancellation is possible. Requiring superlinear terminal convergence for every smooth positive timestep density restores the equivalence, but any prescribed finite collection of such tests can be fooled. A concrete example even has the same second-order terminal coefficient as an exactly acceleration-preserving field while its interior error is first order.

The second answer is a nonlinear topological obstruction. In odd dimension, scalar dilations of arbitrary finite nondegenerate Gaussian mixtures admit no globally Lipschitz acceleration-preserving generator unless the scale is affine. The result allows multimodal, nonradial sources and corrections that redistribute mass across density level sets. Its proof uses a weighted stress identity and the topology of remote level sets; it does not assume a linear field or tangential shell motion. Gaussian consequences include an explicit residual floor at fixed Lipschitz budget and a necessary growth rate for the Jacobian of any hypothetical smooth escape. A convex Gaussian dilation passes every convex-observable minimum-energy test, so this obstruction cannot be recovered by adding more tests of that type.

The retained constructive results supply the positive side: smooth nonradial paths beyond affine-radial/product laws, a Gaussian-start variant, and a globally recoverable correction in a specified finite family. Computable residual and learned-path bounds describe approximation error. The experiments include favorable and unfavorable solver comparisons; no practical speedup or unrestricted correction algorithm is claimed.

## II. Setting and the finite-time distribution bound

Let $p_t \in \mathcal{P}_2(\mathbb{R}^d)$ be generated by the exact flow of $u$. For a grid $0 = t_0 < \cdots < t_N = T$, set $h_k = t_{k+1} - t_k$ and $H = \max_k h_k$. Euler recursively advances $\widehat{X}_{k+1} = \widehat{X}_k + h_k u(t_k, \widehat{X}_k)$ with $\widehat{X}_0 \sim p_0$; write $\widehat{p}_k = \mathrm{Law}(\widehat{X}_k)$. Distribution error uses

$$W_2(\mu,\nu) = \inf_{\gamma\in\Gamma(\mu,\nu)} \left( \int \|x-y\|^2 \,\mathrm{d}\gamma(x,y) \right)^{1/2},$$

whereas common-start particle error pairs numerical and exact trajectories initialized by the same random point. The latter is an upper bound on $W_2$ and can have a different convergence order.

A field $w$ preserves $p$ in the weighted weak sense if $\int \nabla\phi \cdot w \,\mathrm{d}p = 0$ for every $\phi \in C_c^\infty(\mathbb{R}^d)$. For a positive density this is $\nabla\cdot(pw) = 0$. Define $\mathbf{a} = \partial_t u + (Du)u$ and $\mathbf{j} = \partial_t \mathbf{a} + (D\mathbf{a})u$. The *acceleration condition* is zero weighted divergence of $\mathbf{a}_t$ under every $p_t$.

For any $p \in \mathcal{P}_2$, let $\mathcal{G}_p$ be the $L^2(p)$ closure of $\{\nabla\phi : \phi \in C_c^\infty(\mathbb{R}^d)\}$ and let $\Pi_p$ be its orthogonal projection. Its orthogonal complement is exactly the space of weighted-solenoidal fields. We use the standard Wasserstein tangent decomposition [4]; $\varepsilon_t = \|\Pi_{p_t}\mathbf{a}_t\|_{L^2(p_t)}$ is zero precisely under the acceleration condition. This quantity depends on the chosen generator.

**Theorem 1** (Finite-time Euler distribution accuracy)**.** *Let $p_0$ have finite second moment. Suppose $u$ is $C^2$ on a neighborhood of $[0,T]\times\mathbb{R}^d$, globally $L$-Lipschitz in space, and $\sup_t \|u(t,0)\| < \infty$. Suppose each $\mathbf{a}_t$ is globally $L_a$-Lipschitz, satisfies the acceleration condition, and*

$$C_a = \sup_t \|\mathbf{a}_t\|_{L^2(p_t)} < \infty, \qquad C_j = \sup_t \|\mathbf{j}_t\|_{L^2(p_t)} < \infty.$$

*Then on every finite increasing grid,*

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT} C_* \sum_j h_j^3 \le e^{LT} C_* T H^2,$$

$$C_* = \frac{C_j}{6} + \frac{LC_a}{2} + \frac{TL_aC_a}{8}. \tag{1}$$

The proof uses the frozen acceleration flow $\Psi_t^\tau$. Since it preserves $p_t$, Euler applied to $\Psi_t^{h^2/2}X$ has the same output distribution as Euler applied to $X \sim p_t$. The displacement $h^2\mathbf{a}_t(X)/2$ supplied by this shift cancels the leading trajectory defect. Taylor's theorem gives a local $O(h^3)$ coupling error; Euler's Lipschitz factor $1+Lh$ then gives the displayed bound. Appendix XVII contains the complete proof and constants. This is the original finite-time result, retained here; the following sections give a global converse and nonlinear obstructions.

The original second-moment identity is

$$m'' = 0, \qquad M'' = 2\mathbb{E}\|u_t\|^2 \ge 2K_{\min}(t), \tag{2}$$

where $m = \mathbb{E}X_t$, $M = \mathbb{E}\|X_t\|^2$, and $K_{\min}$ is the least instantaneous kinetic energy among path generators. Consequently the action of an admissible field is $[M'(T) - M'(0)]/2$. For a minimum-energy gradient generator $v$, orthogonality gives $\|r_t\|^2_{L^2(p_t)} = M''(t)/2 - \|v_t\|^2_{L^2(p_t)}$ whenever $u = v + r$ is admissible. This budget is an existing consequence, not a new construction. Gradient velocities satisfying all theorem hypotheses have zero acceleration and Euler-exact trajectories (Proposition S27).

## III. Sharp global error and the limits of endpoint tests

Projected modified fields and Duhamel propagation of the density defect are established in recent work [7, 8]. The distinction needed here is between an upper bound that sums the magnitudes of local defects and the norm of their signed, transported sum. The latter gives an exact coefficient for the actual recursive Euler method and a converse to the acceleration condition.

Write $F_{s,t}$ for the exact flow from $s$ to $t$. For $s \le t$, define the transported acceleration and its accumulated defect by

$$A_{s,t} = (DF_{s,t}\,\mathbf{a}_s)\circ F_{t,s},$$
$$B_t^\sigma = \int_0^t \sigma(s) A_{s,t}\,\mathrm{d}s. \tag{3}$$

The integral belongs to $L^2(p_t;\mathbb{R}^d)$. It transports vectors by the derivative of the flow, not merely by composition with the inverse flow.

**Theorem 2** (Exact global Wasserstein coefficient)**.** *Assume the smoothness, growth, spatial Lipschitz, and moment hypotheses of Theorem 1, without imposing the acceleration condition. Consider grids indexed by $n$, with $H_n \to 0$, and scales $\eta_n \downarrow 0$ such that $H_n = O(\eta_n)$. On the $k$th interval set $\sigma_n(s) = h_{n,k}/\eta_n$, and suppose $\sigma_n \rightharpoonup^* \sigma$ in $L^\infty(0,T)$. Let $\widehat{p}_t^n$ be the law of the piecewise linear Euler interpolation. Then*

$$\sup_{0\le t\le T} \left| \frac{W_2(\widehat{p}_t^n, p_t)}{\eta_n} - \frac{1}{2}\|\Pi_{p_t} B_t^\sigma\|_{L^2(p_t)} \right| \longrightarrow 0. \tag{4}$$

*In particular,*

$$\lim_n \frac{\max_k W_2(\widehat{p}_k^n, p_{t_{n,k}})}{\eta_n} = \frac{1}{2}\max_{0\le t\le T} \|\Pi_{p_t} B_t^\sigma\|_{L^2(p_t)}. \tag{5}$$

*The coefficient is continuous in $t$. No density, fourth moment, or regularity of the projected field is assumed.*

The proof first obtains the classical variational error expansion in $L^2(p_0)$, uniformly in time. It then proves a uniform Wasserstein tangent expansion using the frozen-flow lemma, spatial truncation, and the second-moment tail bound. This establishes the coefficient without assuming differentiability of a velocity-to-density map; complete proofs are in Appendix I.

A useful specialization fixes a positive $C^1$ allocation density $\rho$ with $\int_0^T \rho = 1$ and chooses $\int_0^{t_k}\rho = k/N$. Then $\eta_N = 1/N$ and $\sigma_N \to 1/\rho$ uniformly. With $g_s^T = \Pi_{p_T} A_{s,T}$, the terminal coefficient is

$$\lim_{N\to\infty} N W_2(\widehat{p}_T^{N,\rho}, p_T) = \frac{1}{2}\left\| \int_0^T \frac{g_s^T}{\rho(s)}\,\mathrm{d}s \right\|_{L^2(p_T)}. \tag{6}$$

Cancellation occurs inside this norm. An allocation rule based on an integral of nonnegative local magnitudes generally optimizes an upper bound rather than this exact coefficient.

**Theorem 3** (Characterization of second-order distribution accuracy)**.** *Under the same hypotheses, the following are equivalent for a fixed field $u$ on $[0,T]$:*

1) $\nabla \cdot (p_t \mathbf{a}_t) = 0$ *for every* $t$.
2) *There is a constant* $C$ *such that every finite grid satisfies* $\max_k W_2(\widehat{p}_k, p_{t_k}) \le CH^2$.
3) *Along one sequence of uniform grids, with* $h = T/N \to 0$, $\max_k W_2(\widehat{p}_k, p_{kh}) = o(h)$.
4) *For every positive smooth allocation density* $\rho$, $W_2(\widehat{p}_T^{N,\rho}, p_T) = o(N^{-1})$.

*In item 4 it suffices to use the densities in any fixed, arbitrarily small* $C^1$ *neighborhood of the uniform density* $1/T$.

Consequently, a fixed field has a sharp alternative on uniform grids: the acceleration condition gives $O(h^2)$ error at all nodes; if the condition fails, the maximum over nodes is $c_u h + o(h)$ for a strictly positive $c_u$. A first-order terminal coefficient that vanishes by cancellation can always be made positive by an arbitrarily small smooth perturbation of the allocation density. These assertions concern a fixed field as the grid is refined; they do not assert a common positive constant over a changing family of learned fields.

**Corollary 4** (Odd-dimensional location–scale order dichotomy)**.** *Consider a path* $p_t = \mathrm{Law}(m(t) + g(t)Z)$ *covered by Theorem 6. If either* $m$ *or* $g$ *is nonaffine in time, every fixed generator satisfying the hypotheses above has maximum Euler distribution error* $c_u h + o(h)$ *on uniform grids, with* $c_u > 0$. *If both are affine, the unique acceleration-preserving generator is Euler-exact, and every other fixed generator in the stated regularity class has* $\Theta(h)$ *maximum distribution error.*

This applies, in particular, to odd-dimensional expanding Gaussian paths with a nonaffine scale, and to the location–scale paths of finite nondegenerate Gaussian mixtures specified in Theorem 6.

**Proposition 5** (Any finite set of terminal schedules can miss the defect)**.** *Fix* $a > 0$ *and any finite collection of positive smooth allocation densities* $\rho_1, \dots, \rho_m$ *on* $[0,1]$. *There is a smooth field generating the same path* $p_t = \mathcal{N}(0, e^{2at} I_2)$ *as* $a(I+J)x$ *such that the acceleration condition fails, yet*

$$W_2(\widehat{p}_1^{N,\rho_i}, p_1) = O(N^{-2}) \qquad (i = 1, \dots, m).$$

*Its maximum distribution error on uniform grids is* $\Theta(N^{-1})$. *The field can be arbitrarily close to* $a(I+J)x$ *in its time-dependent matrix coefficients and any prescribed finite number of their derivatives.*

The construction is explicit up to a finite homogeneous linear system. Choose a nonzero smooth $f$ with $\|f\|_\infty < a^2$ and $\int_0^1 f/\rho_i = 0$ for every $i$, and put

$$u_t(x) = (aI + b(t)J)x, \qquad b(t) = \sqrt{a^2 - f(t)}. \tag{7}$$

Here $J$ is planar rotation by $\pi/2$. The projected acceleration is exactly $f(t)x$. Finitely many integral constraints leave nonzero choices of $f$; the proof uses disjoint smooth bumps. The same argument can hide the first-order coefficient at any finite set of observation times in addition to the terminal time.

*a) Matching even the terminal second-order coefficient.:* Take $f(t) = \epsilon \cos(2\pi t)$ with $0 < \epsilon < a^2$ in Eq. (7). Uniform Euler has

$$\lim_{N\to\infty} N^2 W_2(\widehat{p}_1^N, p_1) = \frac{2\sqrt{2}\, e^a a^3}{3}, \tag{8}$$

the same coefficient as the acceleration-preserving field $a(I+J)x$. Nevertheless, along $N$ divisible by four,

$$\lim_N N W_2(\widehat{p}_{1/4}^N, p_{1/4}) = \frac{\sqrt{2}\, e^{a/4}\epsilon}{4\pi} > 0.$$

For the arbitrarily small schedule perturbation $\rho_\delta(t) = 1 + \delta\cos(2\pi t)$, $0 < |\delta| < 1$, the terminal error instead satisfies

$$\lim_N N W_2(\widehat{p}_1^{N,\rho_\delta}, p_1) = \frac{\sqrt{2}\, e^a \epsilon}{2|\delta|}\left(\frac{1}{\sqrt{1-\delta^2}} - 1\right) > 0. \tag{9}$$

Thus even an accurate terminal convergence plot, including its leading second-order coefficient, cannot by itself establish acceleration preservation or second-order accuracy throughout the path.

## IV. Odd-dimensional rigidity for nonlinear corrections

Parity obstructions were previously established for affine Gaussian gauges and for tangential fields with deterministic shell radii [7, 8]. Those restrictions do not exclude a nonlinear field that redistributes mass between angles and radii. The next theorem removes these restrictions for a broad class of fixed-shape probability paths. In particular, it applies to arbitrary finite mixtures of nondegenerate Gaussians, including multimodal and nonradial mixtures.

**Theorem 6** (All-generator rigidity in odd dimension)**.** *Let* $d$ *be odd and let* $\rho(z) = Z_\rho^{-1} e^{-V(z)}$ *be a centered positive probability density, with* $V \in C^3(\mathbb{R}^d)$. *Suppose there are* $c, M, R_0 > 0$ *such that, for* $\|z\| \ge R_0$,

$$z \cdot \nabla V(z) \ge c\|z\|^2, \qquad D^2 V(z) \preceq M I_d. \tag{10}$$

*For* $g \in C^4([0,T])$ *positive and* $m \in C^4([0,T];\mathbb{R}^d)$, *prescribe* $p_t = \mathrm{Law}(m(t) + g(t)Z)$, $Z \sim \rho$. *There is a generator satisfying the regularity, growth, and moment assumptions of Theorem 1 and the acceleration condition if and only if both* $m$ *and* $g$ *are affine in time. In that case the generator is unique within this class:*

$$u(t,x) = m'(t) + \frac{g'(t)}{g(t)}(x - m(t)). \tag{11}$$

*Its material acceleration is zero, and Euler is exact on every finite increasing grid. The conclusion allows arbitrary nonlinear, time-dependent corrections; no tangency or shell-preservation condition is imposed on them.*

The source assumptions concern its tails, not log-concavity. For $\rho = \sum_{j=1}^K \pi_j \mathcal{N}(\mu_j, \Sigma_j)$ with $\pi_j > 0$, $\Sigma_j \succ 0$ and $\sum_j \pi_j \mu_j = 0$, let $Q_j = \Sigma_j^{-1}$. Posterior weights $w_j(z)$ give

$$\nabla V = \sum_j w_j Q_j (z - \mu_j),$$
$$D^2 V = \sum_j w_j Q_j - \mathrm{Cov}_w(Q_j(z - \mu_j)).$$

Consequently $D^2 V \preceq \Lambda I$ and $z \cdot \nabla V \ge \lambda\|z\|^2 - B\|z\|$, where $\lambda = \min_j \lambda_{\min}(Q_j) > 0$, $\Lambda = \max_j \|Q_j\|_{\mathrm{op}}$, and $B = \max_j \|Q_j \mu_j\|$. Thus every such mixture meets the theorem's assumptions. Uniformly strongly log-concave sources with bounded Hessian are another subclass.

*a) Why topology applies without assuming tangency.:* After removing the necessarily affine mean, put $\alpha = g'/g$, $\kappa = g''/g$, and $r = u - \alpha x$. At any fixed time write $p_t \propto e^{-V_t}$. Path preservation and acceleration preservation imply the exact identity

$$\operatorname{tr}((Dr)^2) + r^\top D^2 V_t\, r = \kappa\{x \cdot \nabla V_t - d\}. \tag{12}$$

Large level sets of $V_t$ are spheres up to diffeomorphism. The hairy-ball theorem gives a point on each level where $r$ is normal to that level. At this point weighted divergence and a global Lipschitz bound force $\|r\| \le d\,\mathrm{Lip}(r)/\|\nabla V_t\|$. The left side of (12) stays bounded, whereas the right side diverges if $\kappa > 0$. The second-moment identity excludes $\kappa < 0$ and forces $r = 0$ when $\kappa = 0$. Appendix II supplies the details.

**Corollary 7** (The complete convex-observable hierarchy is insufficient)**.** *For every centrally symmetric source with finite required moments and every positive convex scale $g$, the path $\mathrm{Law}(g(t)Z)$ passes all convex-observable and Hessian-weighted minimum-energy tests of Theorem S18. Its covariance also passes (74). Nevertheless, for odd $d$ the Gaussian path $p_t = \mathcal{N}(0, e^{2t} I_d)$ has no acceleration-preserving generator in the globally Lipschitz class.*

Indeed, for a regular convex $\phi$,

$$F_\phi'' = \mathbb{E}[v^\top D^2\phi\, v] + \kappa\mathbb{E}[X \cdot \nabla\phi(X)] \ge K_\phi, \qquad v = \alpha x,$$

because central symmetry and gradient monotonicity make the last expectation nonnegative. The topological obstruction therefore detects infeasibility that the entire convex-test family misses, not just a finite list of moments.

### A. A quantitative obstruction to increasingly accurate gauges

**Theorem 8** (Gaussian residual floor at bounded spatial regularity)**.** *Let $d$ be odd, $p_t = \mathcal{N}(0, g(t)^2 I_d)$, and fix a time with $g''(t) > 0$. Let $u$ be any generator satisfying the hypotheses of Theorem S11, without acceleration preservation. At this time put $\kappa = g''/g$, $L = \mathrm{Lip}(u - (g'/g)x)$, and $A_d = d + \sqrt{d^2 + d - 2}$. Then*

$$\varepsilon_t \ge \frac{g''}{2\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(1 + \max\left\{1, \frac{2A_d L}{\sqrt{\kappa}}\right\}\right)^2\right] > 0. \tag{13}$$

*In particular, on this fixed path no family with uniformly bounded spatial Lipschitz constants can achieve $\varepsilon_t = O(H)$ as $H \downarrow 0$.*

The bound is conservative but applies to every admissible nonlinear gauge. If $\varepsilon_t \le \eta < g'' e^{-2}/(2\sqrt{2\pi})$, it requires

$$L \ge \frac{\sqrt{\kappa}}{2A_d}\left[\sqrt{2\log\left(\frac{g''}{2\sqrt{2\pi}\eta}\right)} - 1\right]. \tag{14}$$

Thus near-cancellation forces a diverging spatial Lipschitz constant. Corollary S12 turns half the right side of (13) into an actual one-step $h^2$ error coefficient lower bound. For the broader source class in Theorem 6, compactness also gives a positive, possibly nonexplicit, residual gap at each fixed Lipschitz bound (Proposition S6). Even outside the globally Lipschitz class, a hypothetical smooth Gaussian solution in odd dimension must have a Jacobian whose maximum on radius $R$ spheres grows at least linearly in $R$; no polynomial correction can satisfy the condition when $g'' > 0$ (Proposition S5).

The dimension distinction is sharp in the radial subclass. In even dimension, if $g'' > 0$ and $J^\top = -J$, $J^2 = -I$, the known field $u = (g'/g)x + \sqrt{g''/g}\, Jx$ preserves every radial scaling path and has distribution-preserving acceleration. A $C^4$ positive scale with $g'' > 0$ supplies the paper's regularity assumptions. Degenerate zeros of $g''$ require separate square-root regularity checks.

## V. A NONRADIAL POSITIVE CONSTRUCTION

The obstruction concerns a specified class of paths. Nontrivial acceleration-preserving paths do exist. A useful sufficient construction starts from a reference law $\rho$, a field $b$ preserving $\rho$, and smooth diffeomorphisms satisfying

$$\begin{aligned} F_{tt}(z) &= DF_t(z)b(z), \\ p_t = (F_t)_\#\rho, &\qquad u_t(F_t z) = F_t'(z). \end{aligned} \tag{15}$$

Then $\mathbf{a}_t(F_t z) = DF_t(z)b(z)$ and $\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t = \int \nabla(\phi \circ F_t) \cdot b \, \mathrm{d}\rho = 0$.

Identify $\mathbb{R}^2$ with $\mathbb{C}$, take $a, \epsilon > 0$ and $\rho = \mathcal{N}(0, I_2)$, and define

$$F_t(z) = e^{(1+i)at} z + \epsilon e^{(1+i)a\sqrt{2}t} z^2 e^{-|z|^2}. \tag{16}$$

If $(4\epsilon/e)e^{a(\sqrt{2}-1)T} < 1$, these maps are uniformly bi-Lipschitz smooth diffeomorphisms and satisfy $F_{tt} = 2a^2 DF_t Jz$. The complete construction theorem and proof (Theorem S7) verify all the global regularity and moment assumptions and exclude every affine-radial and affine-product representation of the target laws. Replacing the second time coefficient by $\epsilon \sinh((1+i)a\sqrt{2}t)$ gives a Gaussian starting law under the stated smallness condition (Corollary S8). These shape-changing paths are not scalar dilations, so there is no conflict with the rigidity theorem.

The path-preserving family

$$u_{\omega,t}(F_t z) = F_t'(z) + (\omega - a)DF_t(z)Jz$$

admits an exact continuum upper certificate for $\|\Pi_{p_t}\mathbf{a}_{u_\omega,t}\|$. Minimizing over a solenoidal acceleration comparison field eliminates one scalar analytically and leaves a coercive quartic in $\omega$. Evaluating all its real cubic stationary points recovers the unique zero $\omega = a$; a positive local minimum demonstrates the limitation of local optimization (Proposition S9). This is a global recovery guarantee within the specified family.

For $u_a$, an explicit phase coupling satisfies

$$W_2(\widehat{p}_N, p_T) \le \|\widehat{X}_N - F_T(e^{-ia^2Th}Z)\|_2 = O(h^2),$$

while the common-start particle error divided by $h$ tends to the strictly positive constant $a^2T\|DF_T(Z)JZ\|_2$ (Theorem S10). The construction provides a positive counterpart to the global characterization and nonlinear obstruction; it is not an inverse solver for arbitrary supplied density paths.

## VI. Certification and target-path error

The local projected-defect formula is established in recent work [7, 8]. In the present Euclidean setting, Corollary S12 proves it under the smoothness, Lipschitz and second-moment assumptions used here, without a density or a regular projected solenoidal flow:

$$\lim_{h\downarrow 0} h^{-2} W_2((\mathrm{Id}+hu_t)_{\#}p_t, p_{t+h}) = \tfrac{1}{2}\varepsilon_t.$$

The new global coefficient retains cancellation between transported defects; a sum of local norms alone cannot identify that coefficient.

For finite error control, Theorem S11 gives explicit constants $D_H = O(1)$ and $B_H = 1 + O(H)$ with

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT}\left(D_H \sum_j h_j^3 + \frac{B_H}{2}\sum_j h_j^2 \varepsilon_{t_j}\right). \tag{17}$$

Its proof follows the full acceleration flow and controls its change of law, rather than assuming that the projected solenoidal component has a regular flow. Node certificates $\varepsilon_{t_j} \le KH$ give second-order accuracy if the remaining constants are uniform. The quantitative odd-dimensional obstruction makes this residual requirement impossible under a fixed Lipschitz budget on its stated Gaussian paths.

Certification requires an upper bound. For $q \in \mathcal{G}_p$ and $w \in \mathcal{G}_p^{\perp}$, orthogonality gives

$$\max\{0, 2\langle a, q\rangle - \|q\|^2\} \le \|\Pi_p a\|^2 \le \|a - w\|^2,$$
$$\|a - w\|^2 - 2\langle a, q\rangle + \|q\|^2 = \|a - w - q\|^2.$$

The complete statements include weighted-Poisson, approximation-error, compact gauge-search, and time-grid certificates. A finite gradient projection norm is a lower bound and cannot by itself certify small residual.

If $u$ is integrated while a different field $v$ generates the intended target $p_t$, Theorem S14 adds initialization error and $\int_0^T \|\Pi_{p_t}(u_t - v_t)\|_2 \, \mathrm{d}t$ to (17), under its explicit mixed-jerk bound. Population excess flow-matching regression loss controls this latter term by Cauchy–Schwarz. It does not alone imply the required regularity or eliminate discretization error. All proofs and precise assumptions are retained in the supplement.

## VII. An exact test of endpoint cancellation

We evaluate the full Gaussian $W_2$ formula for Eq. (7) with $a = 0.4$, $\epsilon = 0.08$, and compare it with the preserving control $b(t) = a$. Both fields generate exactly the same Gaussian path. The grids are uniform or use $\rho(t) = 1 + 0.2\cos(2\pi t)$, with $N = 16, \dots, 4096$ powers of two. Every distance below uses the exact scalar covariance recurrence; there is no empirical optimal-transport estimator or sampling error.

The analytical terminal coefficient on uniform grids is $0.0900163721$ for both fields. For the cosine field, $NW_2$ at time $1/4$ tends to $0.00995003410$ and the maximum-node coefficient is $0.0121776345$. Under the perturbed allocation, its terminal $NW_2$ tends to $0.00870095137 > 0$. At $N = 4096$, the cosine field's uniform terminal error is $5.36463 \times 10^{-9}$, while its maximum-node error is $2.96663 \times 10^{-6}$. The theorem, rather than a fitted slope, establishes the orders and limits.

TABLE I
Observed orders from $N = 2048$ to 4096.

| Field | Grid | Terminal | Maximum node |
|---|---|---|---|
| Cosine | Uniform | 1.9998 | 0.9969 |
| Cosine | Perturbed | 0.9955 | 0.9977 |
| Preserving | Uniform | 1.9998 | 1.9998 |
| Preserving | Perturbed | 1.9998 | 1.9998 |

For numerical verification, an independent $2 \times 2$ covariance propagation agrees with the scalar recurrence within $1.05 \times 10^{-13}$ in absolute covariance. Twelve independent 100-digit reference calculations agree with the stable double-precision $W_2$ evaluations within $1.45 \times 10^{-12}$ relative error. These are arithmetic cross-checks, not interval enclosures. Eight dedicated tests check the field identity, schedule equation, independent covariance recurrence, coefficients, and convergence behavior. The script, CSV files, exact constants, and verification receipts are in `results/global_validation/`; reproduction uses `scripts/run_global_validation.py`.

This controlled example tests the sharp global theorem and the finite-validation limitation. It does not numerically prove the topological obstruction, whose quantification over all nonlinear fields requires the analytical argument.

## VIII. Experiments: discovery, residuals, and cost

We test the nonradial path in Eq. (48) with $a = 0.4$, deformation amplitude $0.18$, and $T = 1$. The global contraction bound is below $0.313$, so every field evaluation uses a convergent inverse-map iteration. The path, correction family, and fitted correction stay fixed as the Euler grid is refined. Results are deterministic population quadratures, except for the separately identified noisy regression experiment.

*a) Correction discovery and certification.:* We eliminate the solenoidal coefficient $c$ from the exact residual formula in Proposition S9, enumerate every real root of the remaining quartic's derivative, and evaluate the original sum of squares at each root. This recovers $(\omega, c) = (0.4, 0.32)$ to floating-point precision without initializing at the solution. The coefficient identity in the proposition certifies the exact zero; the approximately $2.1 \times 10^{-16}$ numerical residual is a floating-point evaluation, not an interval enclosure. A local least-squares run started at $\omega = -0.3$ instead converges to $\omega \simeq -0.392394$, with residual upper bound $0.07241$. Thus a successful local optimization termination does not establish feasibility. Global enumeration takes about $0.01$ seconds including three local comparison runs.

*b) Distribution and particle errors.:* Figure 2 reports the second-moment lower bound and the rotation-coupling upper bound from Theorem S10. We evaluate these continuum bounds by 64-point radial Gauss–Laguerre quadrature and 64 equally spaced angles. They are not exact evaluations of $W_2$, and the quadrature is not interval certified. At $N = 512$, corrected Euler gives bounds $3.439 \times 10^{-7}$ and $4.881 \times 10^{-7}$, with observed orders $1.9983$ and $1.9992$ from $N = 256$ to 512. Its common-start particle error is $6.609 \times 10^{-4}$, with order

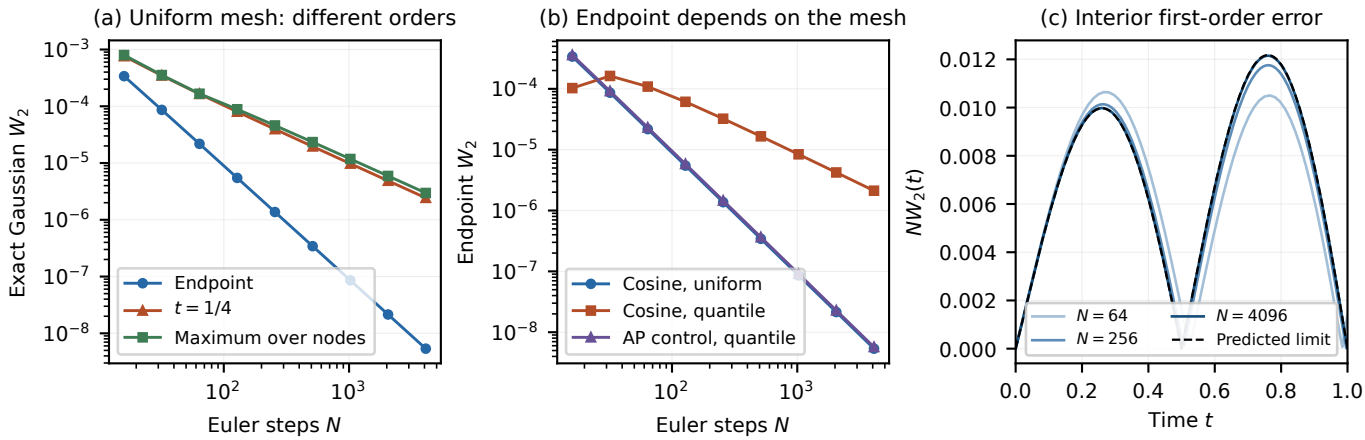



Fig. 1. Endpoint convergence can conceal a first-order path error. The cosine field and the preserving control have the same uniform-grid second-order terminal coefficient. The cosine field has first-order maximum-node error, and a small allocation perturbation makes its terminal error first order as well. Distances are evaluations of exact Gaussian formulas.

TABLE II
NONRADIAL PATH AT EQUAL 384 FIELD EVALUATIONS. THESE ARE QUADRATURE EVALUATIONS OF CONTINUUM $W_2$ BOUNDS.

| Method | $W_2$ lower | $W_2$ upper | Particle RMS |
|---|---|---|---|
| Corrected Euler | $6.111 \times 10^{-7}$ | $8.675 \times 10^{-7}$ | $8.810 \times 10^{-4}$ |
| Base Heun | $6.073 \times 10^{-7}$ | $6.085 \times 10^{-7}$ | $6.085 \times 10^{-7}$ |
| Base RK3 | $1.067 \times 10^{-9}$ | $1.070 \times 10^{-9}$ | $1.070 \times 10^{-9}$ |

0.9992. Base Euler's lower and upper bounds both have order 0.9991. Refining to 96 radial nodes and 128 angles changes the corrected upper bound by less than $8 \times 10^{-17}$. Independent DOP853 trajectories at four intermediate times agree with the analytic flows within $6.5 \times 10^{-12}$ on the tested labels. These checks support the exact path-preservation identity; they do not replace it.

*c) Residuals, infeasibility, and model error.:* Changing $\omega$ by $-2\%$ or $+2\%$ gives exact residual upper certificates 0.01345 and 0.01372 at $T$, and the finest observed upper-bound orders become 1.036 and 0.961. On the infeasible Gaussian path $\Sigma_t = \mathrm{diag}(1 + t, 1 + t + 2t^2)$, the scalar energy budget is positive throughout $[0, 1]$, whereas the first coordinate gives the all-field lower bound $\varepsilon_t \geq [4(1 + t)^{3/2}]^{-1}$. A separate one-coefficient noisy velocity regression over three seeds has analytically known learned paths. Refining Euler reduces its integration error while target error approaches the nonzero continuous-path error. This is a restricted parametric learning check, not a neural generative-model benchmark.

*d) Cost and unfavorable baselines.:* Table II preserves the comparison at 384 field evaluations. Heun's evaluated upper bound is slightly below corrected Euler's evaluated lower bound, providing numerical evidence that Heun is more accurate on this path. RK3 is substantially better. Batch runtimes are comparable, and every stage pays for the inverse map; no sampling speedup is established. The correction improves Euler's distribution order, but it is not the best solver in this experiment. Supplement Section IX gives timings, inverse tolerances, refinement data, optimizer failures, model-error floors, and reproduction commands.

## IX. ESTABLISHED EXAMPLES AND SOLVER TRADE-OFFS

The supplement retains all established Gaussian and nonlinear radial controls, proofs, and numerical receipts. For $p_t = \mathcal{N}(0, e^{2at} I_2)$, replacing $ax$ by $a(I+J)x$ improves Euler's distribution order but doubles the leading common-start particle-error coefficient. At equal NFE, corrected Euler and base-field Heun have exactly identical Gaussian output distributions at their shared nodes (Proposition S28). This example therefore establishes no advantage over a second-order solver.

The shape-changing radial example also has second-order corrected distribution error and first-order particle error (Theorem S24). Its retained mixture calculation increases the particle coefficient by approximately 1.93, while giving only 3.3–4.7% smaller distribution error than base Heun at equal NFE. Corrections remain solver-specific: the Euler rotation worsens Heun's leading Gaussian distribution coefficient, while a different rotation cancels it. Complete Heun/RK3 comparisons, sensitivity formulas, and terminal-cancellation examples remain in the supplement. None establishes a wall-clock speedup; the new nonradial comparison favors Heun and RK3.

## X. RELATION TO PREVIOUS WORK

The closest prior results are the October 2026 analyses of Obzherin and Khrulkov [7, 8]. Their density-level backward-error paper identifies the local projected-defect coefficient, formulates a gauge objective, and derives propagation bounds and timestep allocation rules. Their finite-step geometry paper gives sufficient higher-order conditions, approximate-cancellation budgets, affine reachability, and several nonlinear constructions. The local projection formula and generic gauge mechanism are therefore credited as established. Our global theorem supplies an exact coefficient for recursive Euler under the stated Euclidean $P_2$ hypotheses, together with necessary-and-sufficient maximum-node and schedule-family criteria and a finite-test impossibility result.

Their Gaussian parity theorem is restricted to linear fields [7, Theorem 12]. Their shellwise topological result covers tangential corrections with deterministic output radius and a

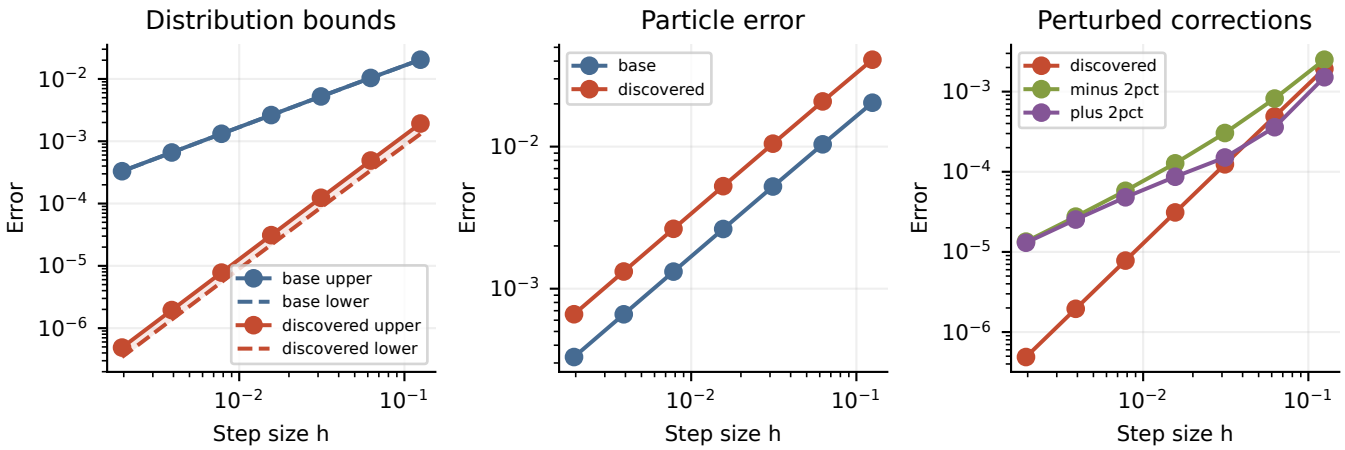



Fig. 2. Nonradial correction discovery followed by fixed-field refinement. Left: quadrature evaluations of lower and upper bounds on full $W_2$, with the region between them shaded. Center: common-start particle RMS. Right: fixed 2% perturbations of the discovered rotation restore an approximately first-order distribution-error trend. The corrected upper bound uses measure-preserving label rotation, not a same-start particle comparison.

positive shell gap [8, Theorem 34]. Our obstruction allows all globally Lipschitz nonlinear corrections and arbitrary finite Gaussian mixtures. Weighted divergence makes a field small at a zero of its tangential projection on a remote density level set; the acceleration identity then gives a contradiction. The topological theorem itself is classical [9]. The extensions are the unrestricted nonlinear obstruction, its residual and regularity consequences, and its separation from every convex-observable energy test.

Gauge freedom, weighted-gradient decompositions, and action minimization are standard [4, 6, 10]. Hurault, Moreau, and Peyré [11] develop propagated weak discretization expansions; the signed transport mechanism is related, while our conclusions concern full $W_2$ and a global converse. Strain and vorticity analyses concern particle-error amplification [12]. Neural Wasserstein Splines projects velocity before differentiation to describe intrinsic curve acceleration [13]; our projected material acceleration belongs to the chosen generator and is representation dependent. The retained reference-coordinate construction supplies positive nonradial examples, while standard solvers, learned-path bounds, and unfavorable comparisons remain explicit. The accompanying literature audit records the two newly identified papers and corrects the first revision's novelty framing.

## XI. Discussion and Limitations

For a fixed regular field, the global theorem gives a complete alternative: acceleration preservation yields second-order maximum-node distribution accuracy; its failure gives a positive first-order coefficient. A terminal plot can conceal that failure, even under finitely many selected schedules. The nonlinear rigidity theorem rules out every admissible correction on its odd-dimensional location–scale paths with nonaffine mean or scale, including multimodal Gaussian mixtures. Convex-observable tests remain useful, but even their entire family does not characterize feasibility.

The global Lipschitz condition is substantive. Gaussian escape from the exact obstruction would require at least linear Jacobian growth on remote spheres, while approximation within a fixed Lipschitz budget has a positive residual gap. These statements do not claim a gap uniform over unbounded regularity budgets. The order dichotomy concerns one fixed field; constants need not be uniform over a changing family. Endpoint cancellation remains possible on infeasible paths and is compatible with the whole-path result.

The nonradial positive construction and its global correction search cover a specified family. They do not solve the inverse problem for arbitrary learned paths. The learned-field experiment is restricted parametric regression, and its theorem requires regularity and mixed-jerk control beyond training loss. The exact Gaussian validation examples and numerical nonradial bounds establish mathematical mechanisms, not an image-generation speedup. Heun and RK3 comparisons retain cases favoring those solvers, and particle errors remain relevant for paired transport and inversion.

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